Sinusoidal Steady-State AC Circuits Answers to the problems

1.1

$$v_0 = 6.154\cos(10^3 t + 70.7°)\text{V}$$

1.2

$$V_0 = 4\sqrt{2}\angle -75°\text{V} \qquad I_0 = 6\sqrt{2}\angle 15°\text{A}$$

1.3

$$i_0 = 0.1 - 0.22\cos(2000t - 45.88°) - 1.18\sin(4000t + 7.4°)$$

1.4

$$R_{ab} = \frac{4}{3}\Omega \qquad R_{cd} = \frac{4}{3}(2 - 3\text{j})\Omega$$

1.5

$$Z_{\text{in}} = 15(1 - \text{j})\text{k}\Omega$$

1.6

$$v_0 = 3.92\cos(1000t - 71.31°)\text{V}$$

1.7

$$P_{4\Omega} = 5\text{W} \qquad P_{2\Omega} = 2.5\text{W} \qquad P_L = P_C = 0\text{W}$$

1.8

$$P_{10\Omega} = 320\text{W}$$

1.9

$$\boldsymbol{S} = 6.614 - \text{j}0.277 \quad \text{kVar}$$

$$\dot{I} = 66.2\angle 92.4°(\text{mA})$$

1.10

$$Z_L = Z_0^* = \frac{1 - \text{j}}{2}, \qquad P_{\max} = 180\text{W}$$

1.11

$$R = |Z_0| = = 2.58\Omega, P_{\max} \approx 7.59\text{W}$$

1.12

$\dot{I}_a = 5.5\angle -45°\text{A}$， $\dot{I}_b = 5.5\angle -165°\text{A}$， $\dot{I}_c = 5.5\angle 75°\text{A}$

1.13

$\dot{I}_a = 44\angle 53.13°\text{A}$, $\dot{I}_b = 44\angle -66.87°\text{A}$， $\dot{I}_c = 44\angle 173.13°\text{A}$